TTS Model Selection Report

Egyptian Arabic text-to-speech — bake-off results

Prepared 23 July 2026 · Egyptian Arabic voice robot stack

Best robot realtime	Best speed	Lowest VRAM	OK / attempted
VoiceTut-TTS	**VoiceTut-TTS**	**VoiceTut-TTS**	**4 / 4**

Executive verdict

Speed/resources default: VoiceTut-TTS (must pass listening). Realtime backup: SILMA-TTS. Chatterbox/NAMAA only if voice quality clearly wins and latency is acceptable.

Overview

4 of 4 TTS models generated audio successfully.
Automated scores cover RTF, throughput (chars/s), VRAM, and load — not voice naturalness.
IMPORTANT: listen to each WAV before choosing a production voice.
RTF = generate_time / audio_length. LOWER better. RTF < 1 = realtime-capable.
Chars/s (HIGHER better). Peak VRAM / RAM / load time (LOWER better).
Top robot realtime pick: VoiceTut-TTS (score 90.02).

How to read this report

TTS — Evaluation methodology & scope

What was evaluated

4/4 models produced WAVs from a shared Egyptian Arabic prompt (~1693 chars).
Outputs saved beside analytics: VoiceTut, SILMA, Chatterbox, NAMAA WAVs.
Hardware reference: NVIDIA T4 ≈ 14.9 GB for co-residency with ASR + LLM.
Speaker/config notes differ per model (see per-model notes in selection report).

How scores were produced

Robot score blends RTF, chars/s throughput, VRAM efficiency, and load (HIGHER better).
Realtime-capable = RTF < 1 (generate time shorter than spoken audio).
Automated ranking intentionally ignores naturalness, accent, and prosody.
Listening checklist on the final page is mandatory for production freeze.

Limitations (read before sign-off)

Naturalness / Egyptian dialect / code-switch quality MUST be judged by ear.
Streaming time-to-first-audio was not measured — validate separately for production.
Scores are relative within these 4 candidates only.
A fast unnatural voice is still a failed production pick.

TTS - Metrics glossary (what to prefer)

Use this page while reading the charts. Direction labels show what you want for a production robot.

Chars/s (characters per second)

HIGHER better
Input characters synthesized per second of compute. Higher = faster speech generation.

Peak CPU %

LOWER better
CPU utilization spike during the run. Lower means less contention with other services.

Generation time (s)

LOWER better
Time to synthesize the full utterance. Lower = robot speaks sooner after the LLM.

Cold load time (s)

LOWER better
Seconds to load the model into memory the first time. Matters for startup / cold starts.

Audio length (s)

Context only
Duration of the produced WAV. Not good/bad alone — used with generate time to get RTF.

RTF (Real-Time Factor)

LOWER better
Processing time / audio duration. RTF < 1 = faster than realtime (good). RTF > 1 = slower than the audio.

Sec per 1k chars

LOWER better
Seconds of compute per 1000 characters. Lower = more efficient synthesis.

x Realtime (xRT)

HIGHER better
How many times faster than realtime (about 1/RTF). Example: 6x = 6s of audio in ~1s of compute.

Realtime capable

Prefer YES
Yes when RTF < 1 (generate time shorter than the spoken audio).

Robot realtime score (0-100)

HIGHER better
Composite score for robot use (speed + resources + accuracy where relevant). Higher = better fit.

Peak VRAM (MB)

LOWER better
GPU memory used at peak. Lower leaves room to run ASR + LLM + TTS together on one GPU.

Balanced score (0-100)

HIGHER better
Normalized blend of the main tradeoffs. Higher = better all-rounder.

Peak RAM (MB)

LOWER better
System (CPU) memory used at peak. Lower is safer on smaller servers / VPS hosts.

Success rate %

HIGHER better
Share of runs that completed without error. Prefer 100% for production reliability.

Remember

For TTS: want LOWER RTF, LOWER generate time, LOWER VRAM/RAM, LOWER load time;
want HIGHER chars/s, HIGHER xRealtime, HIGHER robot score.
Prefer realtime-capable (RTF < 1). Always listen to WAVs — scores ignore naturalness.

Recommended shortlist (listen to confirm)

Best for robot realtime (default)

VoiceTut-TTS

Best composite of RTF + throughput + low VRAM + fast load. Listen: VoiceTut-TTS.wav

score_robot_realtime=90.02 | rtf=0.157 | chars_per_second=82.7 | peak_vram_mb=3150.2

Best realtime alternative

SILMA-TTS

Second on robot score; still realtime (RTF < 1). Use if VoiceTut fails the listen test.

robot=77.4 | rtf=0.238 | vram=3340

Not recommended for live turns

Chatterbox-Multilingual-V3

RTF > 1 (slower than realtime). Consider only if listening quality is clearly superior.

robot=28.6 | rtf=1.101

Not recommended for live turns

NAMAA-Egyptian-TTS

Highest VRAM and RTF > 1. Egyptian-focused candidate— listen, but expect latency cost.

robot=0.6 | rtf=1.232 | vram=6066

Selection notes

Automated picks collapsed to unique roles — VoiceTut swept speed/VRAM/balanced categories.
Naturalness / Egyptian dialect / code-switch quality must be judged by listening to WAVs.
RTF < 1.0 (LOWER better) is preferred for near-real-time synthesis.
For production, also measure time-to-first-audio with streaming APIs.

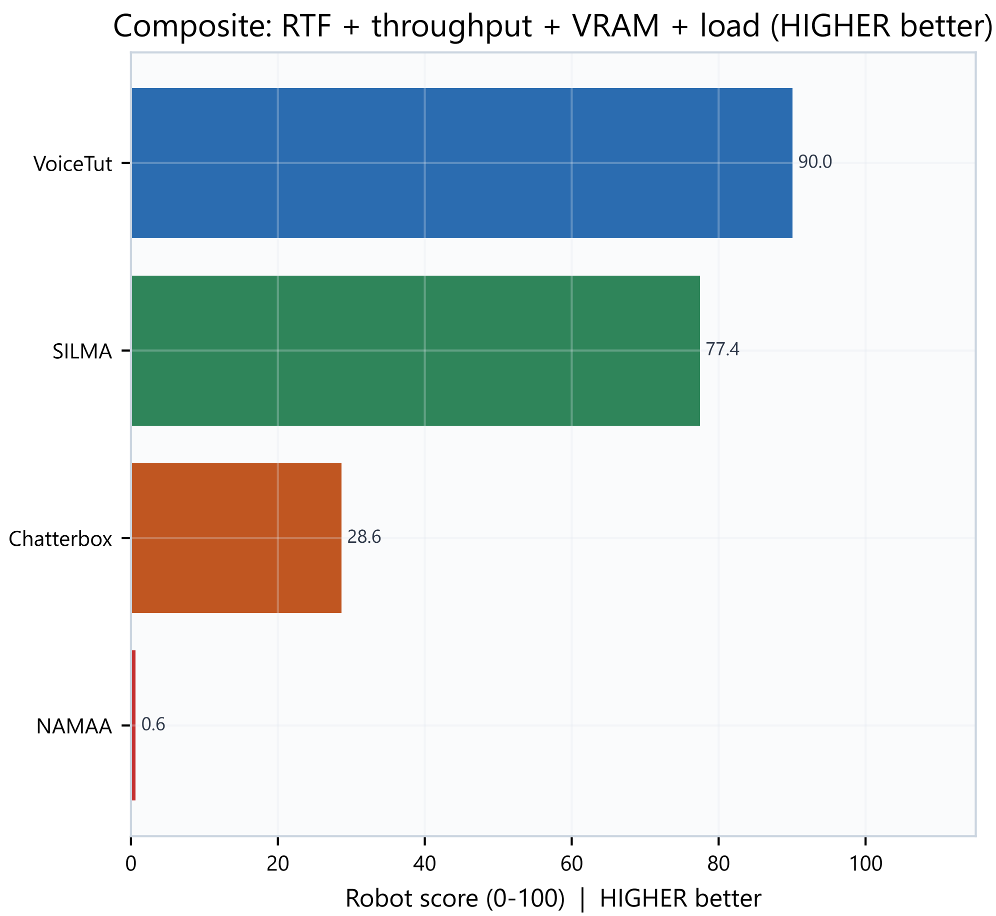
TTS leaderboard (primary evidence table)

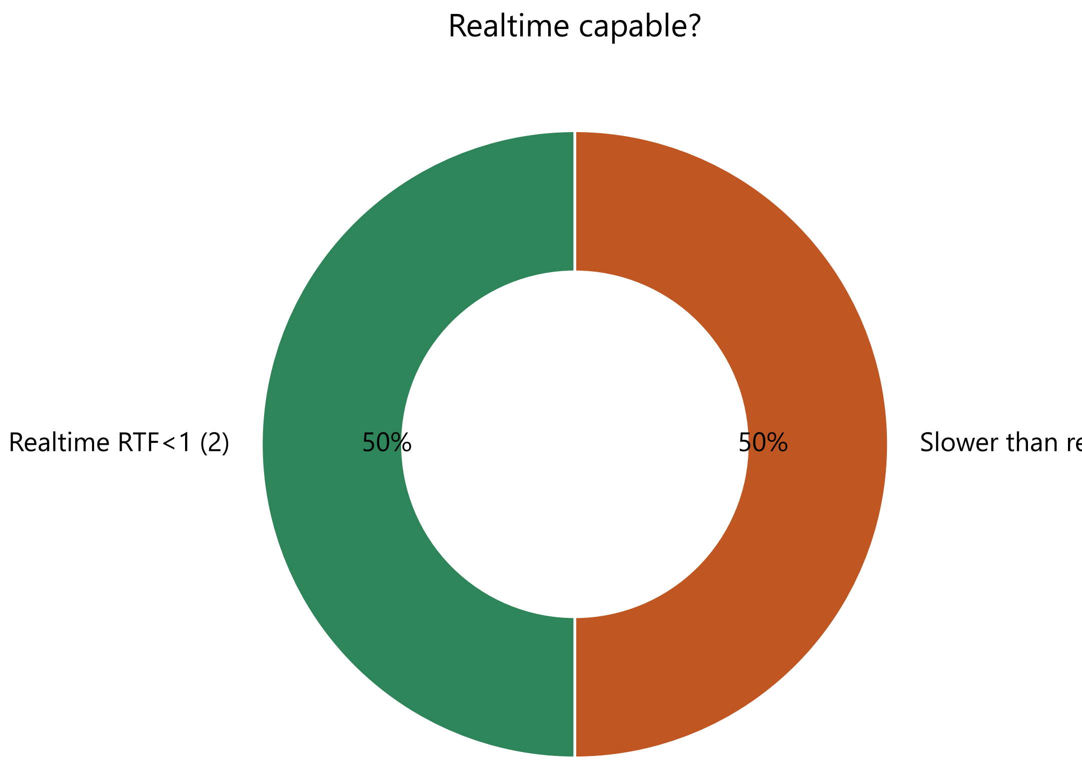
Rank	Model	Robot	RTF	xRT	Chars/s	Gen s	VRAM MB	Realtime
1	VoiceTut	90.0	0.157	6.39	82.7	20.5	3150	yes
2	SILMA	77.4	0.238	4.21	46.8	36.2	3340	yes
3	Chatterbox	28.6	1.101	0.91	13.4	126.3	4790	no
4	NAMAA	0.6	1.232	0.81	15.1	112.5	6066	no

Reading the table

Robot score HIGHER better; RTF/Gen/VRAM LOWER better; Chars/s and xRT HIGHER better.
Realtime = yes when RTF < 1. Only VoiceTut and SILMA qualify in this bake-off.
This table is speed/resources evidence only — listening decides the final voice.
WAV files live in analytics/final/tts_outputs/.

Robot realtime score & Realtime capability

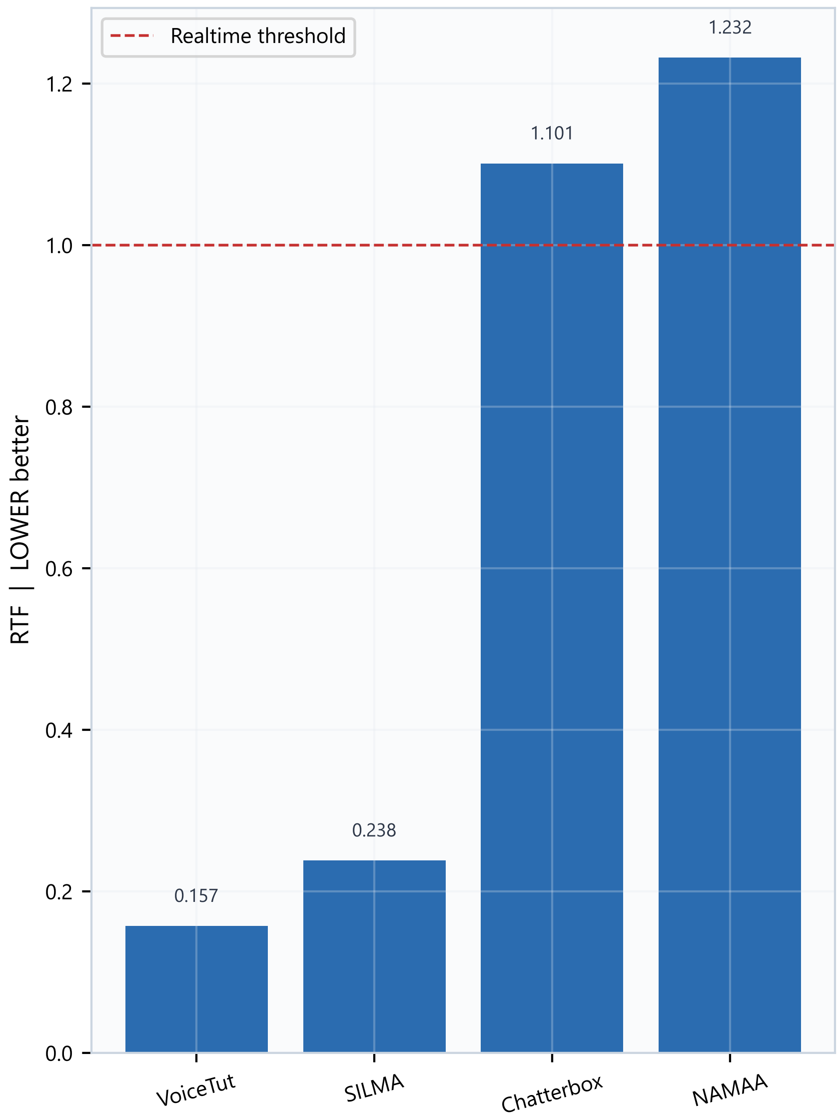


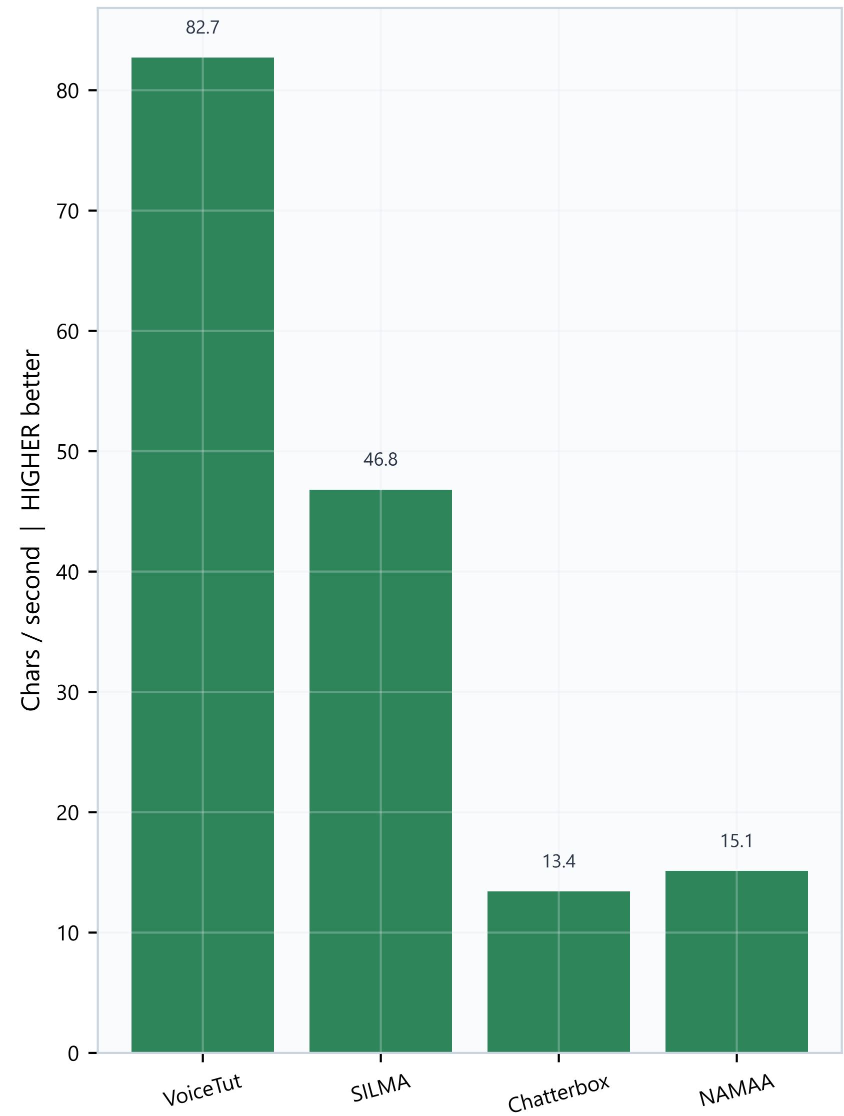


What this means

VoiceTut-TTS leads robot realtime with the best blend of speed and low VRAM.
VoiceTut and SILMA are realtime-capable (RTF < 1); Chatterbox and NAMAA are not.
Non-realtime models can still be used offline or for offline rendering, but hurt live robot replies.
Listen before finalizing — a fast voice that sounds unnatural is still a bad production pick.

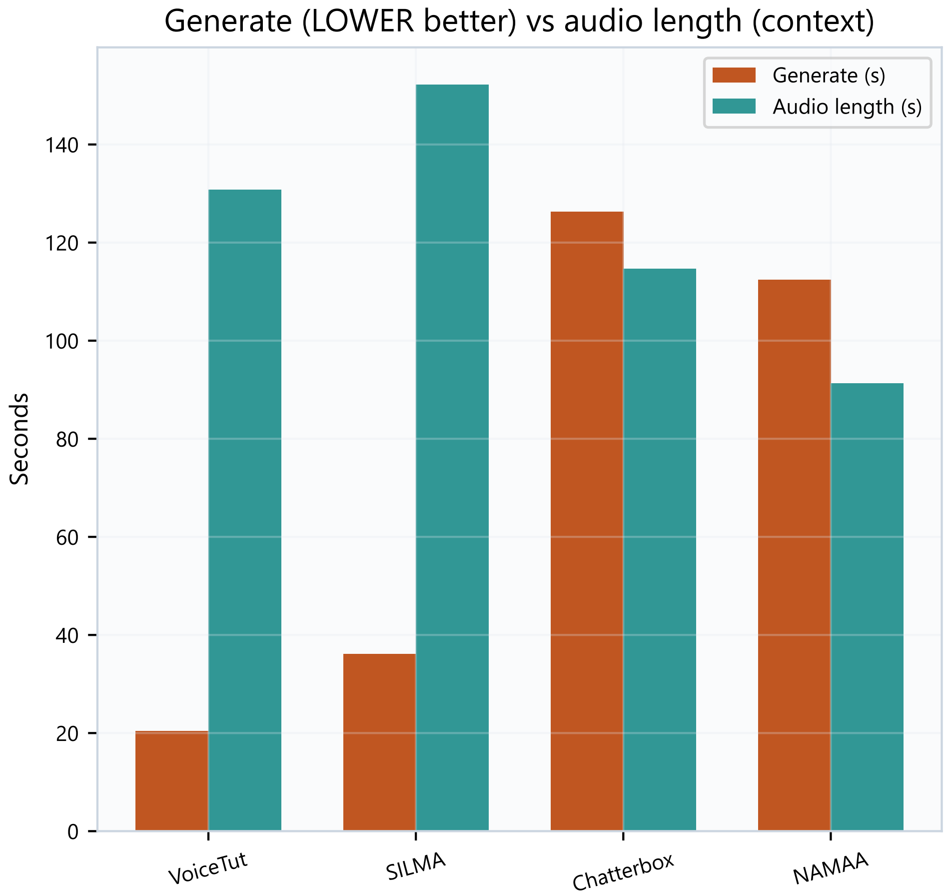
Speed & throughput

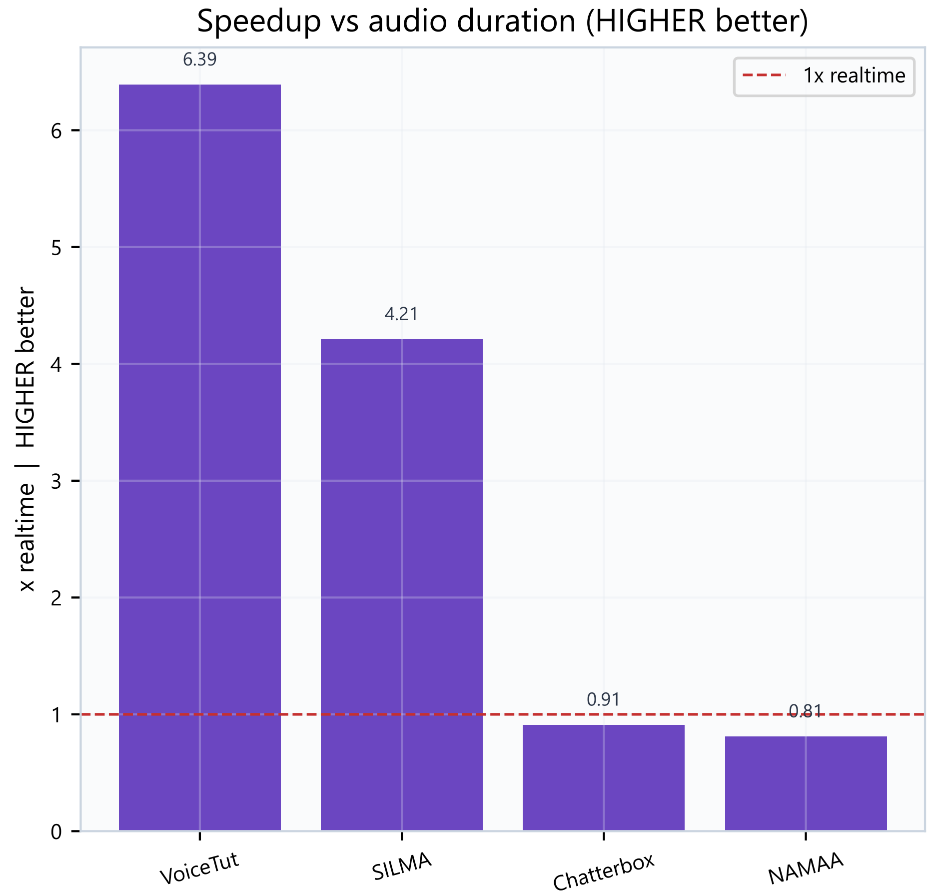




VoiceTut is clearly fastest (RTF 0.157, ~83 chars/s). SILMA is a solid second. Chatterbox and NAMAA take longer than the audio they produce — weak for live robot turns.

Generation time & Speedup

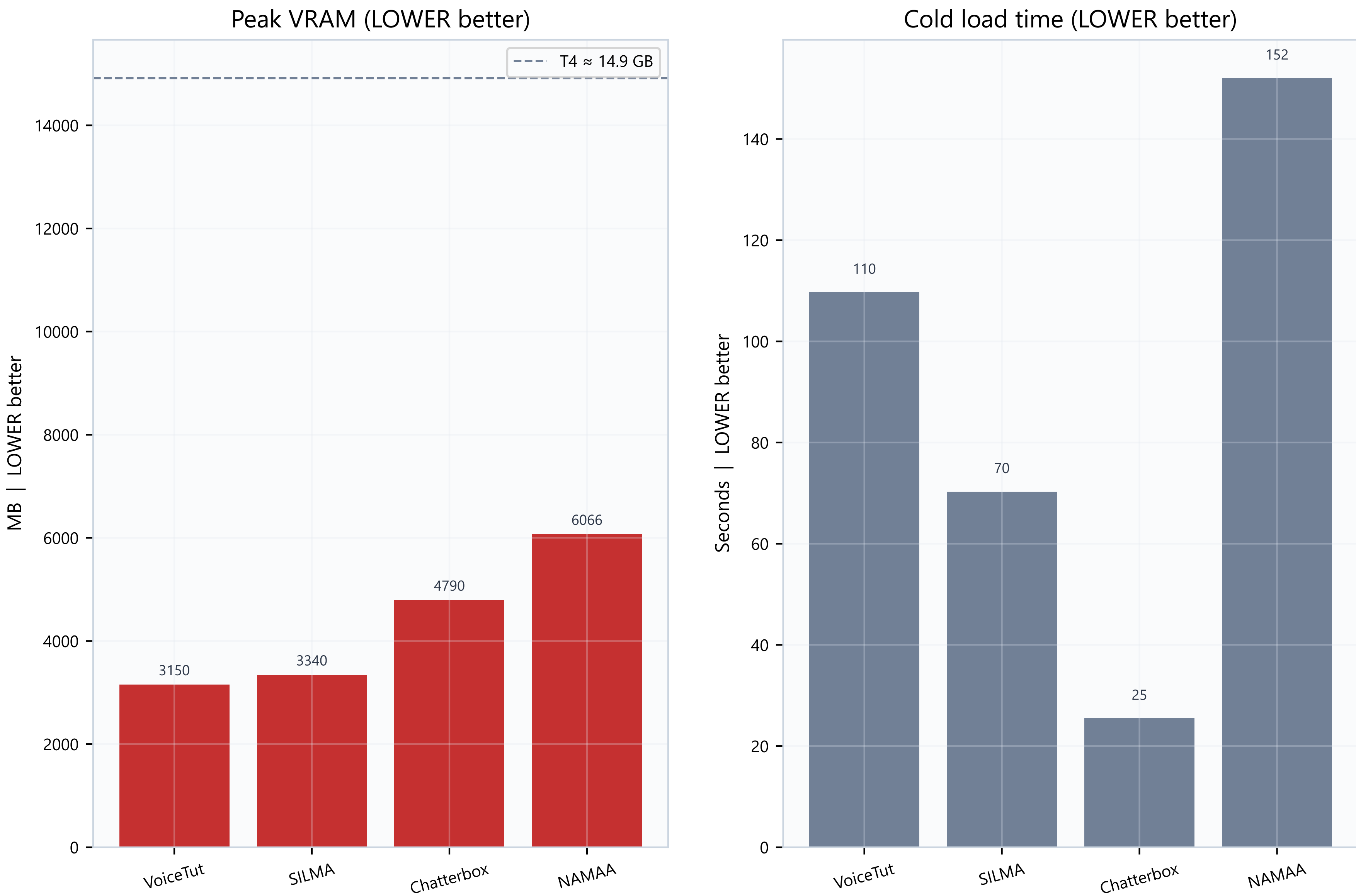




Reading the timing charts

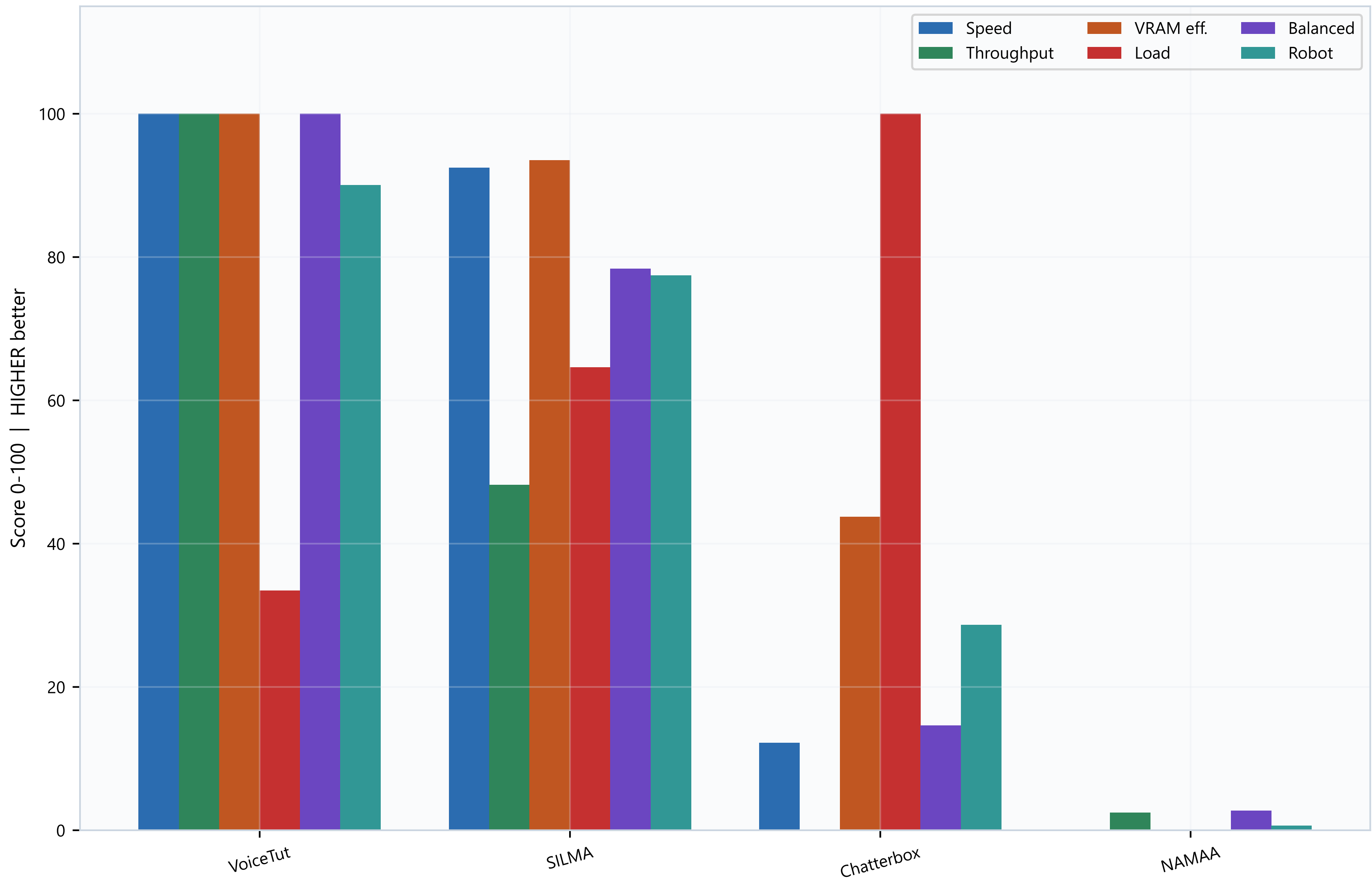
When generate time < audio length, the model is faster than realtime (good).
VoiceTut generates ~131s of audio in ~20s (~6.4x). SILMA ~4.2x.
Chatterbox and NAMAA generate slower than the audio they output.
Cold load time is separate — it hits startup, not every reply, if the model stays warm.

Resources (LOWER better on both charts)



VoiceTut uses the least peak VRAM (~3.15 GB) among OK models — best for co-residency with ASR+LLM. NAMAA is heaviest (~6.1 GB). Chatterbox loads fastest but is not realtime and uses more VRAM.

Normalized score breakdown (all HIGHER better)



VoiceTut sweeps speed/throughput/VRAM/balanced. SILMA is the clear #2. Chatterbox only leads on cold-load score — not enough for live robot use. NAMAA ranks last on robot score.

Listening checklist & how to choose

Listen to these WAVs (score 1–5 each)

1) Egyptian dialect naturalness
2) Arabic / English code-switching
3) Numbers, dates, and times
4) Prosody, pauses, and artifacts
5) Overall robot-voice suitability
Files: VoiceTut-TTS.wav · SILMA-TTS.wav · Chatterbox-Multilingual-V3.wav · NAMAA-Egyptian-TTS.wav

How to choose / sign-off checklist

1) Default speed/resources pick → VoiceTut-TTS (also listen to confirm voice quality).
2) If VoiceTut fails the listen test → try SILMA-TTS next (still realtime, similar VRAM).
3) Prefer Chatterbox/NAMAA only if their voice quality is clearly superior AND latency is acceptable.
4) Confirm combined VRAM with ASR + LLM on the target GPU.
5) For production, also measure streaming time-to-first-audio (not covered in this bake-off).
6) Freeze only after two listeners agree the voice is acceptable for the robot brand.